NBA non-contact injury survival — Update 3: neural survival models

Validation season 2022-23; test season 2023-24 still sealed.

C-index (95% CI)

Cox baseline: 0.764 (0.725–0.799)

DeepSurv: 0.767 (0.728–0.800); vs Cox +0.002 (-0.007 to +0.013)

Transformer + DeepSurv (20 games + all features): 0.774 (0.736–0.806); vs Cox +0.010 (-0.003 to +0.024)

Transformer + DeepSurv (20 games, no workload aggregates): 0.772 (0.734–0.806); vs Cox +0.007 (-0.008 to +0.024)

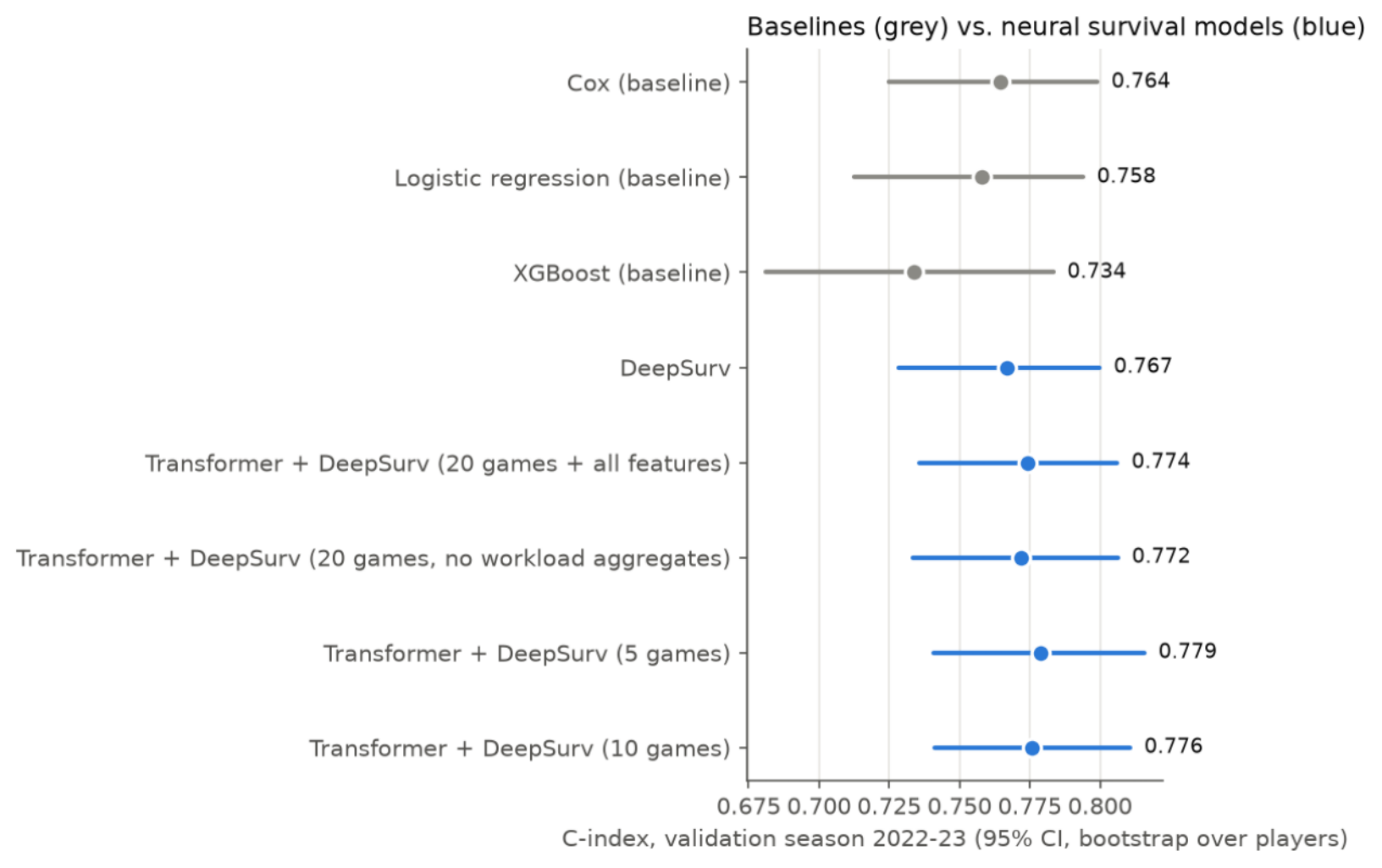
Transformer + DeepSurv (5 games): 0.779 (0.741–0.816); vs Cox +0.015 (+0.003 to +0.029)

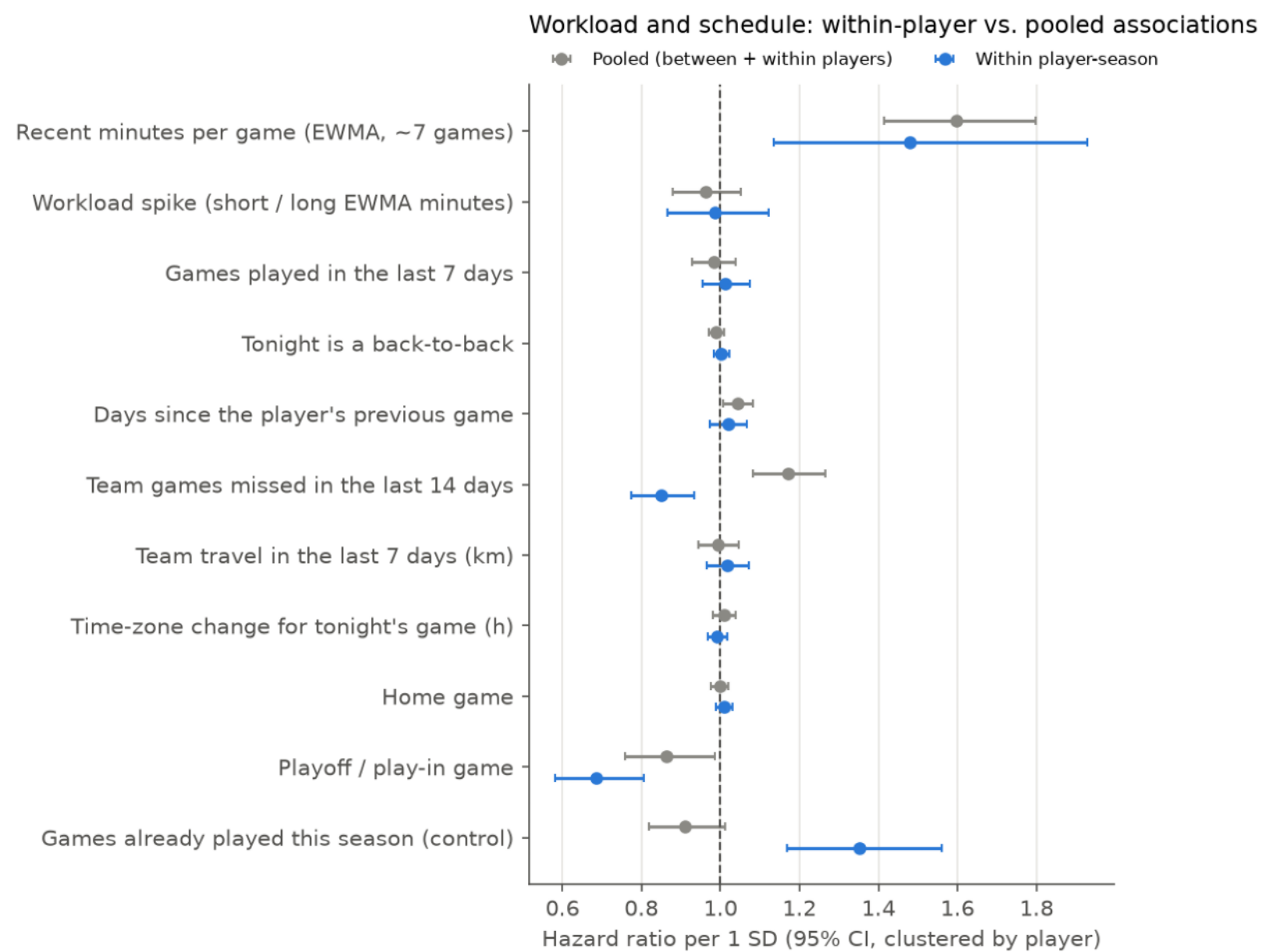
Transformer + DeepSurv (10 games): 0.776 (0.741–0.811); vs Cox +0.011 (-0.001 to +0.025)

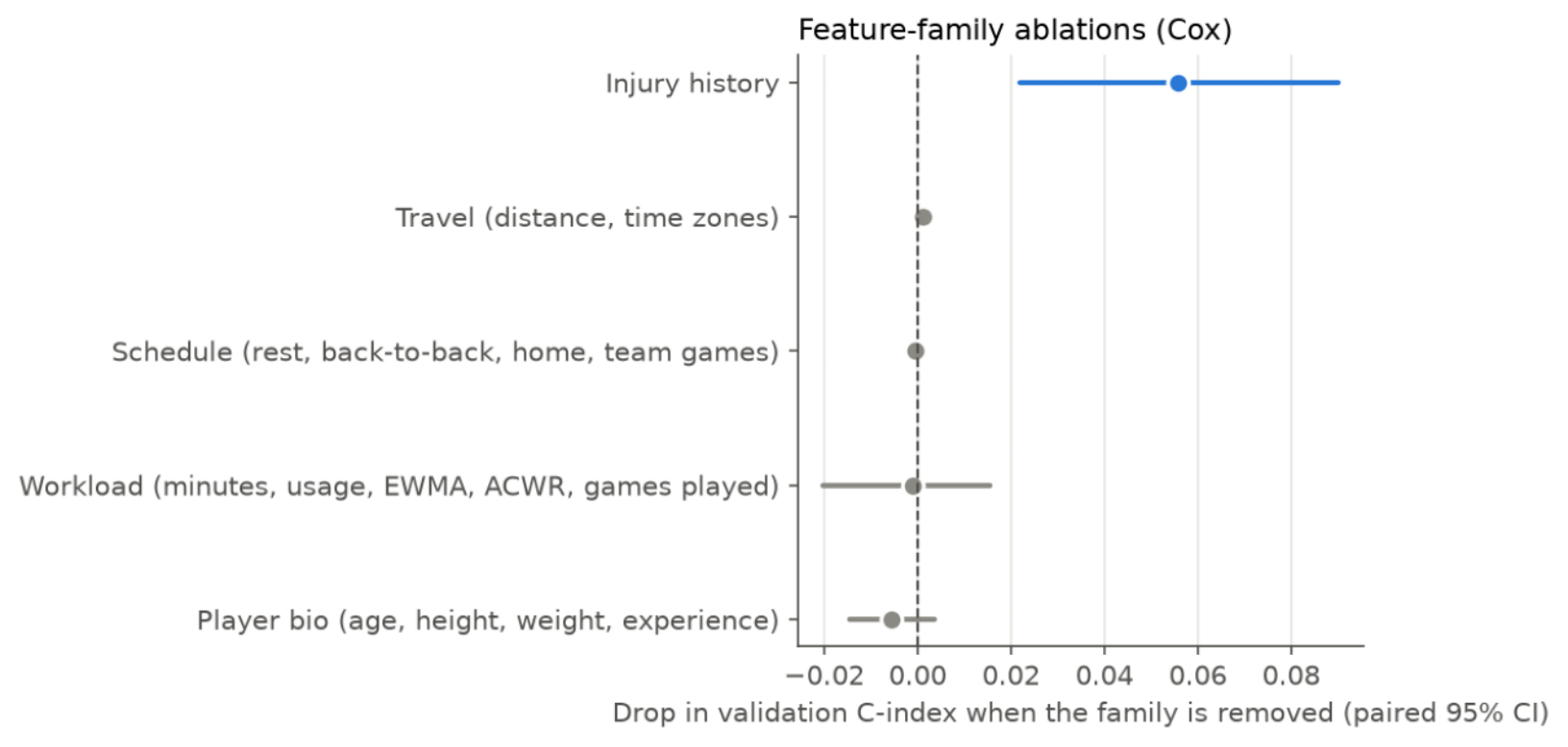
Within player-season, heavier recent minutes precede injury: HR 1.48 per SD (1.13–1.93).

Workload spikes (ACWR), back-to-backs, schedule density and travel: no association.

Ablation: removing history costs the most (C −0.048).


Baselines (grey) vs. neural survival models (blue)
Cox (baseline)
0.764
Logistic regression (baseline)
0.758
XGBoost (baseline)
0.734
DeepSurv
0.767
Transformer + DeepSurv (20 games + all features)
0.774
Transformer + DeepSurv (20 games, no workload aggregates)
0.772
Transformer + DeepSurv (5 games)
0.779
Transformer + DeepSurv (10 games)
0.776
0.675 0.700 0.725 0.750 0.775 0.800
C-index, validation season 2022-23 (95% CI, bootstrap over players)


Workload and schedule: within-player vs. pooled associations
Pooled (between + within players)
Within player-season
Recent minutes per game (EWMA, ~7 games)
Workload spike (short / long EWMA minutes)
Games played in the last 7 days
Tonight is a back-to-back
Days since the player's previous game
Team games missed in the last 14 days
Team travel in the last 7 days (km)
Time-zone change for tonight's game (h)
Home game
Playoff / play-in game
Games already played this season (control)
0.6
0.8
1.0
1.2
1.4
1.6
1.8
Hazard ratio per 1 SD (95% CI, clustered by player)


Feature-family ablations (Cox)
Injury history
Travel (distance, time zones)
Schedule (rest, back-to-back, home, team games)
Workload (minutes, usage, EWMA, ACWR, games played)
Player bio (age, height, weight, experience)
−0.02
0.00
0.02
0.04
0.06
0.08
Drop in validation C-index when the family is removed (paired 95% CI)